MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS
FALL SEMESTER 2025
STUDY GUIDE FOR TEST 1

Chapter 1: Real Numbers

(1) Examples and non-examples of ordered fields
(2) How to deduce some properties of feilds (or ordered fields) from given axioms? (You do not have to memorize the axioms, axioms will be given if relevant questions are asked.)
(3) Definition of supremum and infimum. Relations of supremum and infimum
(4) How to find supremum and infimum of some sets? How to find supremum and infimum (if possible) for some operations e.g. addition, multiplication?
(5) Completeness property and how to use this axiom to show existence of some solution of an equation
(6) The definition of real numbers
(7) Explain why $\mathbb{Q}$ fails the completeness property
(8) Well-orderness property of $\mathbb{N}$
(9) Denseness property of set of rational numbers
(10) State and prove Nested Interval Property. Examples of nested intervals (empty set, one element, more than one elements, etc)
(11) How to show countability of $\mathbb{Q}$?
(12) How to show the set of $\mathbb{R}$ or irrational numbers is uncountable?
(13) How to show some number e.g. $\sqrt{2}$ is not a rational number in $\mathbb{R}$?

Chapter 2: Sequences (up to Part 3 Section 3.3)

(1) Definition of limits ($\epsilon - N$-definitions). Use $\epsilon - N$-definitions to find some some limits
(2) Examples of convergent sequences and divergent sequences. Operations on convergent and divergent sequences
(3) Methods of showing a sequence convergent: Boundedness, Monotone Convergence Theorem, Cauchy sequences, squeeze theorem, use of some known limits e.g. $\lim_{n\to\infty} \sqrt[n]{n}$, $\lim_{n\to\infty}(1+\frac{1}{n})^n$, $\lim_{n\to\infty} r^n$, $\lim_{n\to\infty} \frac{\sin(n)}{n}$.
(4) How to show Monotone Convergence Theorem from definitions.
(5) Define Cauchy sequences. Relation to Convergent sequences. How to show a convergent sequence is Cauchy, and its converse?
(6) Contractive sequence is Cauchy, and examples.
(7) Subsequence criteria for convergence of a sequence

(8) State Monotone Subsequence theorem and Bolzano-Weierstrass Theorem. How to use Monotone subsequence theorem to show Bolzano-Weierstrass Theorem. Find some monotone subsequences of some sequences.
(9) Define a series. Examples of convergent series.
(10) Absolute and conditional convergence, and examples. Relations of absolute convergence, conditonal convergence and convergence, and how to show the relations.
(11) Rearrangement of sequences e.g. for absolutely convergent sequences and conditionally convergent sequences. Examples on rearrangement of sequences. Rearrangement Theorem.
(12) Tests for convergence: n-th term tests, comparison thests and limit comparison tests.